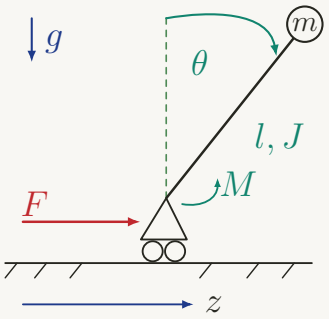
EXERCISE 05

Transfer Functions

RECAP

Bigger picture

1) We want to control a system...



Inverted pendulum

So we find the DE's:

$$\dot{x} = f(x, u)$$
$$y = g(x, u)$$

- $x = (z \;\; \dot{z} \;\; \theta \;\; \dot{\theta})^T$
- $u = F$
- $y = \theta$

2) Linearize around an equilibrium point...

- Find Jacobi-matrices and plug in.

$$x_e = (0 \;\; 0 \;\; 0 \;\; 0)^T, \quad u_e = 0$$

$$\begin{pmatrix} \dot{x} \\ \ddot{x} \\ \dot{\theta} \\ \ddot{\theta} \end{pmatrix} = \underbrace{\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}}_{A} \begin{pmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{pmatrix} + \underbrace{\begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}}_{B} u$$

$$y = \underbrace{(c_1 \;\; \cdots \;\; c_n)}_{C} \begin{pmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{pmatrix} + \underbrace{d}_{D} \cdot u$$

3) With A, B, C, D and the initial condition of the states x_0, we can find $y(t)$ using...

$$y(t) = \underbrace{Ce^{At}x_0}_{\substack{\text{Initial condition}\\ \text{response}}} + \underbrace{\int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau + Du(t)}_{\substack{\text{Forced}\\ \text{response}}}$$

This is where we left off. We simplified the problem using diagonalization of the A matrix and can now tackle control... Or can we???

1 | Two problems of the time domain

1) The integral in the solution

$$y(t) = Ce^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau + Du(t)$$

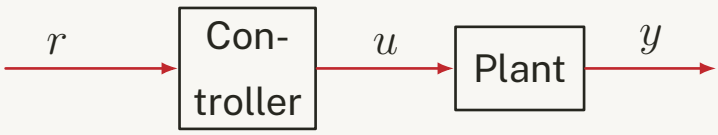
$\Rightarrow$ This is actually a **convolution**!

$$\text{DEF } (f * g) = \int_0^t f(\tau)\,g(t-\tau)\,d\tau$$

$\Rightarrow$ Convolutions are computationally intense.

2) Connecting signals

- There will be multiple blocks!



$\Rightarrow$ Input u to the plant is the output of the controller, so to find y, we must evaluate

$$u(t) = C_1e^{A_1t}x_0 + \int_0^t e^{A_1(t-\tau)}B_1\,r(\tau)\,d\tau + D_1\,r(t),$$

and only then we can find

$$y(t) = C_2e^{A_2t}x_0 + \int_0^t e^{A_2(t-\tau)}B_2\,u(\tau)\,d\tau + D_2\,u(t).$$

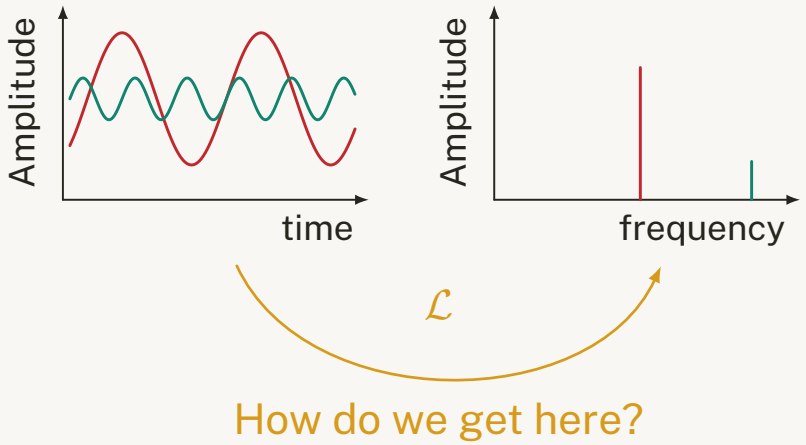
$\Rightarrow$ This is not very efficient, and there is a very clever way to tackle these problems.

Things will get a little abstract now, it's okay if you don't immediately have an intuition.

2 | Frequency domain

- GENERAL IDEA Instead of looking how a system behaves over **time**, we look at how it behaves over different **frequencies**.

- This means we represent signals differently:



$\Rightarrow$ We use the **Laplace-transform**:

$$\mathcal{L}\{f\}(s) = \int_0^\infty f(t)\, e^{-st}\, dt$$

- We only need **one** transformation pair, the exponential:

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

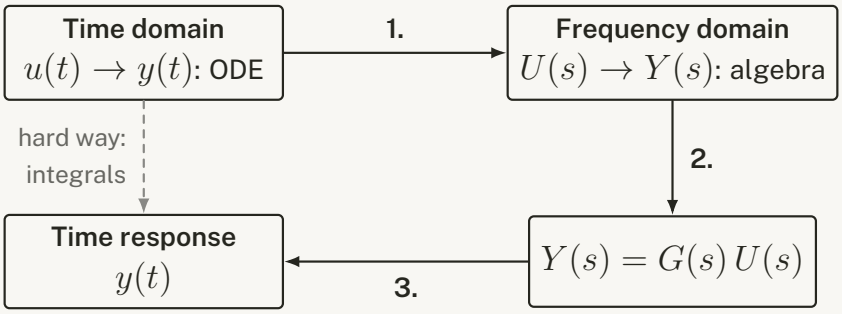
⇒ Every input can be built from **exponentials** e^{st} with a complex number s, so this pair is enough.

- The Laplace transform also has the following identity:

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s)$$

⇒ A derivative in time becomes a **multiplication** with s.

- This gives us the whole process:



1. **Transform** the problem from time to frequency domain with $\mathcal{L}$.

2. **Solve** it there: only algebra, e.g. $Y(s) = G(s)\,U(s)$.

3. **Transform back** to the time domain with $\mathcal{L}^{-1}$ to get the time response $y(t)$.

⇒ Instead of solving a differential equation, we only have to do **algebra**!

$$\dot{x}(t) = Ax(t) + Bu(t) \qquad \text{(I)}$$

$$y(t) = Cx(t) + Du(t) \qquad \text{(II)}$$

⇒ Apply $\mathcal{L}\{\}$ to (I):

$$\mathcal{L}\{\dot{x}(t)\} = \mathcal{L}\{Ax(t) + Bu(t)\}$$

$$sX(s) = AX(s) + BU(s)$$

$$(s\mathbb{I} - A)X(s) = BU(s)$$

$$X(s) = (s\mathbb{I} - A)^{-1}BU(s) \quad \text{(III)}$$

⇒ Now transform (II):

$$\mathcal{L}\{y(t)\} = \mathcal{L}\{Cx(t) + Du(t)\}$$

$$Y(s) = CX(s) + DU(s) \quad \text{(IV)}$$

⇒ plug (III) into (IV):

$$Y(s) = C(s\mathbb{I} - A)^{-1}BU(s) + DU(s)$$

$$Y(s) = \big(C(s\mathbb{I} - A)^{-1}B + D\big)U(s)$$

- From this, we can define the next, very important concept:

The **Transfer Function**.

$$G(s) = \frac{Y(s)}{U(s)} = C(s\mathbb{I} - A)^{-1}B + D$$

⇒ It describes the ratio of **output** to **input** of a system in frequency domain!

Let's try an example...

3 | Example — Exam HS2022

Problem: You are given a linear time-invariant system of the form:

$$\dot{x}(t) = A \cdot x(t) + B \cdot u(t),$$
$$y(t) = C \cdot x(t) + D \cdot u(t).$$

} state-space ⇒ time domain

Q16 (1 Points)

Given:

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}, \; B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \; C = \begin{bmatrix} 0 & 1 \end{bmatrix}, \; D = -1.$$

Which of the following transfer functions $G(s)$ is equivalent to the given state space system. Mark the correct answer.

- ☐ $G(s) = -\frac{s^2-3\cdot s+2}{s^2-4\cdot s+1}$
- ☐ $G(s) = -\frac{(s+2)\cdot(s-1)}{s^2-4\cdot s+1}$
- ☐ $G(s) = \frac{s^2-3\cdot s+2}{s^2-4\cdot s+1}$
- ☐ $G(s) = -\frac{s^2+3\cdot s+2}{s^2-4\cdot s+1}$

} transfer-functions ⇒ frequency domain

APPLY the formula $G(s) = C(s\mathbb{I} - A)^{-1}B + D$:

$$G(s) = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} s-1 & 1 \\ 2 & s-3 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ -1 \end{pmatrix} - 1$$

- The inverse can be time-intense...

⇒ Use this identity:

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)}$$

$$\text{With adj} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

$$\text{and det} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb.$$

$$\Rightarrow \begin{pmatrix} s-1 & 1 \\ 2 & s-3 \end{pmatrix}^{-1} = \frac{1}{(s-1)(s-3)-2} \begin{pmatrix} s-3 & -1 \\ -2 & s-1 \end{pmatrix}$$

$$G(s) = \frac{1}{(s-1)(s-3)-2} \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} s-3 & -1 \\ -2 & s-1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} - 1$$

$$= \frac{1}{(s-1)(s-3)-2} \begin{pmatrix} -2 & s-1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} - 1$$

$$= \frac{-s-1}{(s-1)(s-3)-2} - 1 = \frac{-s-1-(s-1)(s-3)+2}{(s-1)(s-3)-2}$$

$$= \frac{-s^2+3s-2}{s^2-4s+1} = -\frac{s^2-3s+2}{s^2-4s+1}$$

- ☒ $G(s) = -\frac{s^2-3\cdot s+2}{s^2-4\cdot s+1}$
- ☐ $G(s) = -\frac{(s+2)\cdot(s-1)}{s^2-4\cdot s+1}$
- ☐ $G(s) = \frac{s^2-3\cdot s+2}{s^2-4\cdot s+1}$
- ☐ $G(s) = -\frac{s^2+3\cdot s+2}{s^2-4\cdot s+1}$

1P!

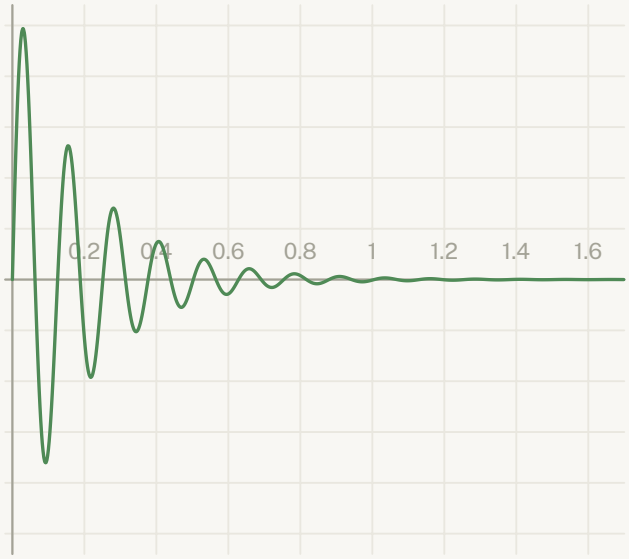
4 | What are transfer functions and s really?

So we now know how to get to the frequency domain. But what really is a transfer function? And what really is "s"?

Sinusoids and exponentials

- The purpose of the Laplace-Transform...

⇒ The Laplace-Transform describes a function as a **combination** of **sinusoids** and **exponentials** only.



$$f(t) = e^{-5t}\sin(50t)$$

We can describe many signals in this form!

⇒ Visualization

`https://johnflux.com/2019/02/12/laplace-transform-visualized/`

A general signal

- So lets look at a general signal:

$$f(t) = \lambda e^{\sigma t}\cos(\omega t)$$

- Rewrite using $\underbrace{\cos(x) = \frac{1}{2}\left(e^{jx} + e^{-jx}\right)}_{\text{Euler's Formula}}$

$$\Rightarrow f(t) = \lambda\frac{1}{2}e^{\sigma t}\left(e^{j\omega t} + e^{-j\omega t}\right)$$

$$f(t) = \lambda\frac{1}{2}\left(e^{(\sigma+j\omega)t} + e^{(\sigma-j\omega)t}\right)$$

- The signal is fully characterized by the exponent $\sigma \pm j\omega$, and the magnitude λ.

- Due to linearity, we can now write general signals as

$$f_{\text{gen}}(t) = \sum_i \lambda_i e^{(\sigma_i \pm j\omega_i)t}$$

Let's take a closer look at $\sigma + j\omega$.

The complex frequency s

- $\sigma + j\omega$ has the structure of a complex number $a + jb$. The complex frequency is named:

$$s = \sigma + j\omega$$

$\Rightarrow$ $\text{Re}(s) = \sigma$: **exponential** part

$\text{Im}(s) = \omega$: **sinusoidal** part

- If we now want to know what a system $G(s)$ does with some complex frequency, we just plug in...

EX. Given a pure sinusoid with $\omega = 1 \frac{\text{rad}}{\text{s}}$, what does $G(s)$ do with it?

$$G(s) = \frac{1}{s^2 + 2s + 2}$$

Plug in $s = \sigma + j\omega$

$s = 1j \frac{\text{rad}}{\text{s}}$

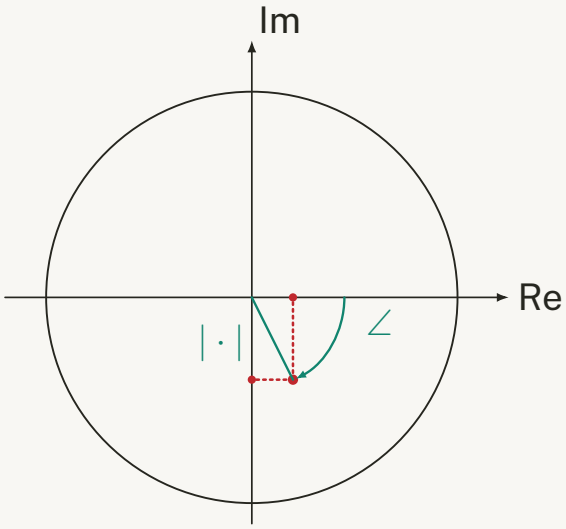
$$= \frac{1}{-1 + 2j + 2} = \frac{1}{1 + 2j} = \frac{1 - 2j}{5} = \frac{1}{5} - \frac{2}{5}j$$

Amplitude and phase

So we just get another complex number?

⇒ Yes! From which we can extract the **amplitude** and the **phase**.

$G(s) = \frac{1}{5} - \frac{2}{5}j$

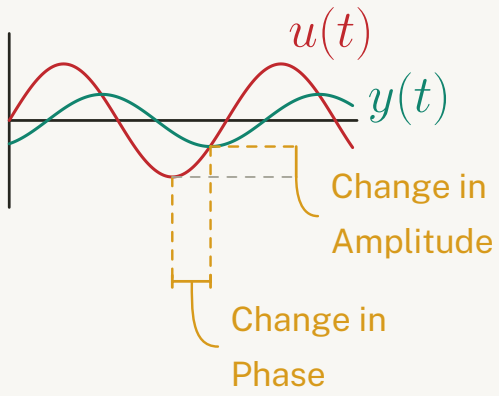


$|\cdot|$: Magnitude

$\angle$: Phase

- Magnitude: $\left|\frac{1}{5} - \frac{2}{5}j\right| = \sqrt{\left(\frac{1}{5}\right)^2 + \left(-\frac{2}{5}\right)^2} = \frac{\sqrt{5}}{5} \approx 0{,}45$
- Phase: $\angle\left(\frac{1}{5} - \frac{2}{5}j\right) = \arctan\left(\frac{-\frac{2}{5}}{\frac{1}{5}}\right) \approx -1{,}11\,\text{rad}$

A seemingly random number and some angle? What does this mean?



- Remember, our $G(s)$ is a **fraction**!

$$G(s) = \frac{Y(s)}{U(s)} = \frac{1}{s^2 + 2s + 2}$$

$\Rightarrow$ It shows us the ratio of output to input.

- $|G(s)|$: ratio of amplitudes
- $\angle G(s)$: phase change

Let's recap...

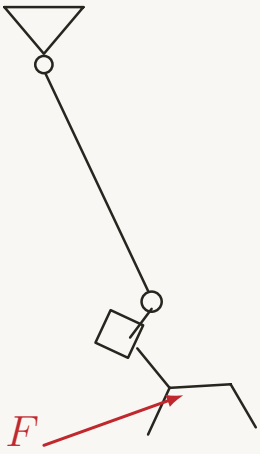
Excitation frequency

- Many systems are excited with sinusoidal frequencies. With a transfer function, we find out how the system **warps** the amplitude and phase of a system.

$\Rightarrow$ And the **frequency**?

- The output of a **LTI**-system, always has the same frequency as the input!

INTUITION



- $F_1(t) = \cos(1000\,t)$
- $F_2(t) = \cos(0{,}001\,t)$

$\Rightarrow$ You must hit the right frequency to strongly **amplify** the amplitude.

https://www.youtube.com/watch?v=LV_UuzEznHs

5 | Convolution theorem and connecting blocks

Now we start to see the use of the frequency domain.

⇒ But theres more!

1) The integral in the solution

$y(t) = Ce^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau + Du(t)$

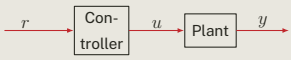
⇒ This is actually a convolution!

Def: $(f * g) = \int_0^t f(\tau)g(t-\tau)d\tau$

⇒ Convolutions are computationally intense

2) Connecting signals

- There will be multiple blocks!



- These are the problems we still had in time domain. The Laplace transform has one more identity that solves these issues!

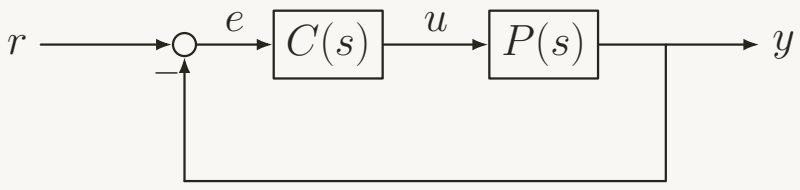
- CONVOLUTION THEOREM

$$\mathcal{L}\{f(t) * g(t)\} = F(s) \cdot G(s)$$

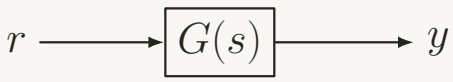
⇒ So the hard to solve **integral** becomes a simple **multiplication**.

- Also, if we have a connection of blocks, we can evaluate the total system by simply multiplying the transfer functions.

EX. $C(s) = \dfrac{1}{s+2}, \quad P(s) = \dfrac{s+2}{(s-1)(s+1)}$



we already saw, that we can rewrite:



where $G(s) = \dfrac{C(s)P(s)}{1 + C(s)P(s)}$

so here $G(s) = \dfrac{\frac{1}{s+2} \cdot \frac{s+2}{(s-1)(s+1)}}{1 + \frac{1}{s+2} \cdot \frac{s+2}{(s-1)(s+1)}}$

$$= \frac{\frac{1}{\frac{(s-1)(s+1)}{(s-1)(s+1)}}}{\frac{(s-1)(s+1)+1}{(s-1)(s+1)}} = \frac{1}{s^2} \;//$$

Entering frequency domain is really useful. But what if I have a tf given and need to go back to time-domain?

6 | Back to the time domain: realization

- We can find one possible realization with this formula:

GIVEN

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0} + d$$

THEN

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & & & \ddots & 1 \\ -a_0 & -a_1 & & \cdots & & -a_{n-1} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

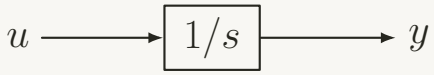
$$C = \begin{bmatrix} b_0 & b_1 & \cdots & b_{n-1} \end{bmatrix}, \qquad D = \begin{bmatrix} d \end{bmatrix}$$

$\Rightarrow$ This realization is called the **controllable canonical form**

That's all the theory for today. Let's look at some quiz questions...

QUIZ

Q1) What does the following tf do?



Answer: $y = \int u\,dt$

$$\left.\begin{aligned}\dot{x} &= u\\ y &= x\end{aligned}\right\} \; y(t) = x(t) = \int u(t)\,dt$$

$\Rightarrow$ write as $\dot{x} = ax + bu,\; y = cx + du$

$\Rightarrow\; a = 0,\; b = 1,\; c = 1,\; d = 0$

Find TF as: $G(s) = C(s\mathbb{I} - A)^{-1}B$

$$= 1 \cdot (s - 0)^{-1} \cdot 1 = \underline{\underline{1/s}}$$

Q2) The transfer function of any system can be written as a ratio of two polynomials in s.

True / **False**

$\Rightarrow$ Must be SISO-LTI!

Q3) Exam HS22

Q17 (1 Points)

Given:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -2 & 0 & -4 & -1 & -6 \end{bmatrix}, \; B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \; C = \begin{bmatrix} \frac{1}{2} & -2 & 0 & -\frac{1}{3} & 0 \end{bmatrix}, \; D = 0.$$

Which of the following transfer functions $G(s)$ is equivalent to the given state space system. Mark the correct answer.

☐ $G(s) = \dfrac{\frac{1}{2} \cdot s^3 - 2 \cdot s + \frac{1}{3}}{s^5 + 2 \cdot s^4 + 4 \cdot s^3 + 1 \cdot s^2 + 6}$

☒ $G(s) = \dfrac{-\frac{1}{3} \cdot s^3 - 2 \cdot s + \frac{1}{2}}{s^5 + 6 \cdot s^4 + s^3 + 4 \cdot s^2 + 2}$

☐ $G(s) = \dfrac{-\frac{1}{3} \cdot s^3 - 2 \cdot s + \frac{1}{2}}{s \cdot (s^4 + 6 \cdot s^3 + s^2 + 4 \cdot s + 2)}$

☐ $G(s) = \dfrac{s \cdot (\frac{1}{2} \cdot s^3 - 2 \cdot s + \frac{1}{3})}{s^5 + 2 \cdot s^4 + 4 \cdot s^3 + 1 \cdot s^2 + 6}$

GIVEN

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0} + d$$

THEN

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & & & \ddots & 1 \\ -a_0 & -a_1 & & \cdots & & -a_{n-1} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} b_0 & b_1 & \cdots & b_{n-1} \end{bmatrix}, \qquad D = \begin{bmatrix} d \end{bmatrix}$$